PyGameMaker — Tutorial 3: Classic Pong — Teacher Guide

Companion for the in-app tutorial (**Help > Tutorials > Classic Pong: Two-Player Game**, 4 pages) and the matching student handout and worksheet. Students should have finished Tutorial 2 (events, actions, collisions, score, draw).

Overview

Students build Pong in three phases: paddles and a bouncing ball, invisible goals, and a score display. New ideas: **two players on one keyboard**, **solid objects and bouncing**, **global variables** (global.p1score), and drawing a variable's value.

> Pong is a good moment to talk about *rules*. Every rule is an event plus actions on one object: "when the ball touches the left goal, add 1 to Player 2's score and put the ball back".

Suggested Timing (45-60 minutes)

The tutorial says 20-25 minutes for a confident student; in class allow more.

Segment	Time	What happens
Recap and predict	5 min	Ask: "What objects does Pong need?" Let students list them before starting
Phase 1: paddles and ball	15-20 min	Pages 1-2; the longest phase, many objects
Phase 2: goals and scoring	10 min	Page 3
Phase 3: score display	5-10 min	Page 4
Play and swap	5 min	Pairs play each other; then swap computers
Worksheet / exit ticket	5 min	Worksheet parts A-C

For a 45-minute slot, do Phases 1-2 in class, Phase 3 as homework.

Step-by-Step Walkthrough and Common Issues

Phase 1: paddles and ball

- Students create six objects here. Suggest a checklist on the board: wall, two paddles, ball, then later two goals and a score object.
- **Solid** matters. Walls and paddles must be Solid so that *Bounce against solid objects* has something to bounce off. The ball is *not* Solid.
- Each paddle needs its own **Collision with obj_wall** event with *Stop movement*. Without it, a solid wall does not stop a paddle that has no collision event for it.
- The ball needs a Bounce event **for each object** it should bounce off (wall, left paddle, right paddle). Students often add only the wall.
- In this phase, the ball leaves the screen at the sides; that is **expected**.
- If students have the ball moving only sideways: the direction field of *Start moving in direction* must be 45 (up and to the right, 0 is right and 90 is up).

Phase 2: goals and scoring

- Goals are **invisible** but **Solid**: invisible so players don't see them, present so the ball's collision event fires.
- The score variables are **global**: global.p1score, global.p2score. A typo in one place (for example global.p1Score) creates a second, different variable, and the score seems stuck at 0.
- Left goal scores for **Player 2**, right goal for **Player 1** (the player who did not miss).
- After a goal the ball is returned to its start position but keeps its direction, so it heads for the same goal again. This is normal; it makes a good challenge ("reverse the ball's direction after each goal").

Phase 3: score display

- obj_score has no sprite and must be placed in the room, like the controller in Tutorial 2.
- Draw text is **black by default** and a new room's background is black, so without *Set draw color* the score is invisible. The tutorial page includes that block. The text is placed at y 40 and 60, below the 32-pixel wall row, so it does not overlap the top wall.

> **Reference projects.** Download the checkpoint projects from the wiki (03_pong_checkpoints.zip): one project for the end of each phase. Give a stuck student the previous phase so they can continue.

Vocabulary Introduced

Term	What it means here
Solid	Objects that block or bounce others (walls, paddles)
Bounce	An action that reverses the direction of movement after a collision
Global variable	A named value shared by all objects; always written with global.
Draw event	Runs every frame; used to show text and numbers
Start position	Where an object was placed in the room

Discussion Questions

- Which objects does Pong need, and which are invisible?
- Why do we keep the score in a global variable?
- What would happen if the goals were not Solid?
- Is the game fair? (The ball keeps its direction after a goal.) How could we fix it?

Differentiation

- **Support:** give the Phase 1 checkpoint and start at Phase 2; pair a stronger and a weaker student.
- **Extension:** speed the ball up on each paddle hit; end the game at 10 points; change the bounce angle by where the ball hits the paddle.

Worksheet Answer Key

Part A: 1-D, 2-C, 3-A, 4-B.

Part B: 1. obj_paddle_left. 2. obj_ball. 3. The ball (its Collision with a goal event adds to the score). 4. obj_score. 5. The paddle's Collision with obj_wall event.

Part C:

1. A collision event belongs to one specific other object; the game only runs the bounce for objects that have an event.
2. The left goal is behind Player 1's paddle; when the ball gets there, Player 1 missed, so Player 2 scores.
3. The score has to survive the ball being reset, and several objects (the ball adds to it, the score object draws it) use it, so it lives in a shared global variable.

Parts D and E: completion and reflection.

Rubric: Pong

Level	What the game shows
4 - Complete	Both paddles move and are held by the walls; the ball bounces off walls and paddles; goals score for the right player and reset the ball; both scores are displayed
3 - Working	Paddles, ball and scoring work; the score display or one bounce is missing
2 - Partly there	Paddles move and the ball bounces, but goals do not score
1 - Started	Some objects exist, but the ball does not bounce or a paddle does not move

End-of-Session Checklist

- [ ] Every student has a saved project (Ctrl+S)
- [ ] Every pair has played at least one point
- [ ] Note who needs the Phase 1 checkpoint next time